

Using likelihood ratio functions for modeling the conditional probability of occurrence of future landslides for risk assessment

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Abstract

The most crucial and difficult task in landslide hazard analysis is estimating the conditional probability of occurrence of future landslides in a study area within a specific time period, given specific geomorphic and topographic features. This task can be addressed with a mathematical model that estimates the required conditional probability in two stages: “relative hazard mapping” and “empirical probability estimation.” The first stage divides the study area into a number of “prediction” classes according to their relative likelihood of occurrence of future landslides, based on the geomorphic and topographic data. Each prediction class represents a relative level of hazard with respect to other prediction classes. The number of classes depends on the quantity and quality of input data. Several quantitative models have been developed and tested for use in this stage; the objective is to delineate typical settings in which future landslides are likely to occur. In this stage, problems related to different degrees of resolution in the input data layers are resolved. The second stage is to empirically estimate the conditional probability of landslide occurrence in each prediction class by a cross-validation technique. The basic strategy is to divide past occurrences of landslides into two groups, a “modeling group” and a “validation group”. The first mapping stage is repeated, but the prediction is limited to only those landslide occurrences in the modeling group that are used to construct a new set of prediction classes. The new set of prediction classes is compared to the distribution of landslide occurrences in the validation group. Statistics from the comparison provide a quantitative measure of the conditional probability of occurrence of future landslides.

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1. Introduction

Geomorphologists traditionally have constructed landslide hazard maps by identifying areas likely to be affected by future landslides, based on geomorphic features seen on aerial photographs and noted during field surveys (Panizza et al., 1998). Civil engineers have constructed slope stability

maps, using deterministic models of mass movement and debris flows based on slope angles, soil cohesion, water saturation capacities, shearing resistance and other physical properties. Each location in the resulting stability map is assigned a level of “safety factor of slope failure” for the unit area surrounding the location (Terlien et al., 1995; Soeters and van Westen, 1996).

Hazard maps constructed by geomorphologists typically show 3–5 discrete levels of relative hazard.

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Slope stability maps produced by civil engineers show the safety factor as a continuous scale of measurement. These prediction maps representing both landslide hazard and slope stability have a common purpose—to guide decision-makers in land-use planning. However, land-use planners face the problem of how to interpret the mapped hazard levels. For example, if a relatively large subarea has been assigned to a specific “hazard class” in a hazard map or has consistently “high” values of safety factor in a slope stability map, it may be difficult to make a decision on what to do about the subarea. A ban on development in the subarea would certainly avoid possible future damage that might result from landslides, but the economic consequences of precluding many types of development could have a serious socioeconomic impact over the large subarea. In other words, a decision maker must balance the economic consequences of a loss from a landslide against the economic cost of foregone opportunities.

What decision makers need from hazard maps or slope stability maps are not only relative levels of landslide hazard but also estimates of the probability of occurrence of future landslides at any given location. If estimates of probabilities of occurrence could be made, then cost-benefit analysis could be used and decision-makers could quantitatively compare the economic consequences of a ban on development against the possible loss that would result if a landslide occurred. Decision-makers could arrive at a learned and informed decision, rather than one based on emotions or “gut-feelings.”

Although several quantitative prediction methods have been proposed (Cardinali et al., 2002; Carrara, et al., 1995; and Corominas et al., 1998), these have not discussed the estimation of conditional probability of occurrence of landslides. A two-stage approach has been suggested by Chung and Fabbri (2002, 2003, and 2004) and is summarized here with mathematical details, for estimating the probability of occurrence of future landslides given a spatial database of geomorphic and topographic data from a sub-area under the assumptions of a scenario. The first stage is to construct a hazard map with a number of hazard classes, similar to what is done by geomorphologists or civil engineers. The second stage is to estimate the probability that a landslide may occur in the areas defined by each class under a given scenario. Depending on the locations and timings of past landslides, it may not be possible to

estimate probabilities of occurrence of future landslides under certain scenarios for which insufficient data are available.

The basic strategy is that past landslides in the study area are divided into two mutually exclusive time periods. Occurrences in the earlier time period are used to construct a prediction model that is used to generate a prediction map with several classes of landslide hazard. Counting landslides from the later time period that fall in the mapped hazard classes predicted from the earlier time period yields an estimate of the conditional probability of occurrence of landslides in the hazard classes in the future. If a time division is not possible because there is no information on the dates of past landslides, then the landslides can be randomly or spatially divided into two groups. In such a random division or spatial division, however, the two divisions do not provide a natural extension of the estimated probabilities to occurrences of future landslides.

Consider a study area with m spatial data layers. The m data layers may include thematic data layers and continuous data layers; both types of data are believed to correlate spatially and genetically with landslides. A thematic data layer of a property such as bedrock geology consists of an array of pixels, each of which is assigned an integer value that uniquely represents one of the mutually exclusive thematic classes. In a continuous data layer, each pixel represents a physical property such as elevation, and the pixel array consists of continuous values. These m values at each pixel location are used to construct a prediction model that measures the relative hazard of future landslides at the pixel location. Several favorability function models (Chung and Fabbri, 1993, 1998, 1999, 2001, and 2004) have been developed to construct predictions.

The prediction methodology will be illustrated using the likelihood ratio function as the favorability function model. Several difficult problems in the simultaneous use of thematic and continuous data layers and the use of input data layers having different spatial resolutions or map scales also are addressed. A Canadian case study from the La Baie area in the Province of Québec (Perrét, 2004, personal communication) is used as an example only to illustrate the spatial modeling procedure and no serious attempt was made to generate a spatial prediction model for the locations of future landslides in the case study.

2. Input spatial database and the case study

Let A denote a study area with m data layers of geomorphologic and topographic information related to the occurrences of landslides in A . In addition to the m data layers, there is a landslide data layer containing the scars of past landslides in A . In some instances the landslide scars may be small with respect to the pixel resolution or map scale; the locations of such past landslides are recorded as points on the map rather than polygons. In some studies, as of debris flows, geomorphologists may delineate scarps where landslides have been triggered. If the geomorphic and topographic characteristics of scarps differ significantly from characteristics of scars, it is imperative that the scarps rather than the scars be studied and modeled for landslide hazard mapping. Depending on the type of landslides, when referring to a “landslide” in hazard mapping, we mean the point location or a polygon of a scar or the scarp of a landslide.

The first step in a quantitative study is to overlay a fine grid over A , such that each grid cell represents a small unit area. The geographic extent of a unit area is dependent both on the map scales of the original input documents and on the purpose of the study. Grid cells typically range from 5×5 to 50×50 m on the ground. Each grid cell is referred to as a “pixel,” because it appears as a single pixel when the grid is displayed on a computer screen. In each data layer a single value, termed a “pixel value,” is assigned to each pixel. For the landslide data layer, a “1” is assigned to any pixel located within a past landslide and a “0” is assigned to any pixel that is outside the areas of past landslides. A digital elevation model (DEM) is used to generate three types of spatial information for each pixel; (1) slope angle, (2) aspect angle, and (3) elevation. All three measures usually are included in an analysis as separate continuous data layers.

The La Baie study area (Québec, Canada) is 10×5 km in extent. The data layers representing the study area contain 2000×1000 pixels; each pixel represents a 5×5 m area on the ground. There are five layers of geomorphic and topographic information related to landslides, including (1) bedrock geology (12 rock types), (2) forest coverage (binary), (3) elevations, (4) aspect angles and (5) slope angles. The locations of 25 landslides that occurred in 1964 and 64 landslides that occurred after 1964 are mapped. Twelve of the more recent landslides are reactivations of landslides that occurred in 1964.

The average size of the 89 landslides is approximately 15×15 m, or 9 pixels. Because of the close proximity of some landslides, scars of the 25 landslides of 1964 are represented by 22 polygons and scars of the 64 post-1964 landslides are represented by 55 polygons in the database; all 89 landslides form only 67 separate polygons. Of the 2,000,000 pixels in the data set, 223,593 pixels represent a lake and rivers and were excluded from the analysis; the study includes the remaining 1,776,407 pixels.

3. Favorability function model for relative-hazard mapping

Consider a pixel x in A with m pixel values, $(x_1, \dots, x_m)$, one for each map-layer, and let $Y(x)$ represent the presence ($Y(x) = 1$) or absence ($Y(x) = 0$) of a landslide at pixel x . Following Chung and Fabbri (2003, 2004), suppose that a “favorability” function $g(x : x_1, \dots, x_m)$ at a pixel $x \in A$, is constructed as a real function such that:

1. $g(x : x_1, \dots, x_m)$ represents (or measures) a relative level of hazard of pixel x for given m pixel values, $(x_1, \dots, x_m)$ for the occurrence of a landslides ($Y(x) = 1$).
2. Consider two pixels, a and b with values, $(a_1, \dots, a_m)$ and $(b_1, \dots, b_m)$. If the m pixel values $(a_1, \dots, a_m)$ for a th pixel provide a more hazardous geomorphic and topographic environment for landslides than $(b_1, \dots, b_m)$ for the b th pixel, then:

$$g(a : a_1, \dots, a_m) \geq g(b : b_1, \dots, b_m).$$
3. The function can be estimated using data contained in the database.

If a function could be constructed at every pixel in A , a relative hazard map in A could be created by computing $g(x : x_1, \dots, x_m)$ at every pixel $x \in A$. In addition to the three essential attributes of favorability functions described above, desirable properties should include:

4. Simultaneous analysis of continuous and thematic data layers without converting one type of data into another. Most prediction techniques use either thematic data layers (Clerici et al., 2002) or continuous data layers (Carrara et al., 1995; Guzzetti et al., 1999) but not both.

5. Ability to handle different resolutions (or scales) of data layers simultaneously, when different resolutions are contained in a data base.

There are several existing mathematical functions that can be used as favorability functions as discussed by (Chung and Fabbri (1993, 1998, 1999, 2001, and 2004)). The likelihood ratio function is used in this example.

4. Likelihood ratio function model

To simplify the discussion and assess the role of an individual data layer as a predictor of landslides, consider the use of a slope map calculated from a DEM. Suppose a study area is divided into two non-overlapping subareas, a portion containing landslides and the remainder. Suppose further that slope angles provide useful information for identifying landslides, and that slope angles in the landslide areas should have unique characteristics different from slope angles in the remaining areas. Frequency distributions of scars and the remaining areas should be distinctly different. The likelihood function is the ratio of the two frequency distributions and highlights the difference between the two areas; it also is a favorability function that satisfies the five conditions listed above.

A prediction model identifying areas of future landslide occurrences can be generated based on the likelihood ratio function of the multivariate frequency distributions calculated for these layers. To formalize this idea, consider a point x with m pixel values $(x_1, \dots, x_m)$ in a study area A that consists of two subareas, that containing landslides M and a remaining landslide-free area $\bar{M}$. Let $f\{x_1, \dots, x_m|M\}$ and $f\{x_1, \dots, x_m|\bar{M}\}$ be the multivariate frequency distribution functions assuming that a pixel is from M and from $\bar{M}$, respectively. Then the *likelihood ratio* (Kshirsagar, 1972; Press, 1972; Cacoullos, 1973) at x is defined as:

$$\lambda(x : x_1, \dots, x_m) = \frac{f\{x_1, \dots, x_m|M\}}{f\{x_1, \dots, x_m|\bar{M}\}}. \quad (1)$$

Estimate $\lambda(x : x_1, \dots, x_m)$ in Eq. (1) using, for example, discriminant analysis or empirical methods (see Chung, 2003; Chung and Keating, 2002; Chung et al., 2002a, for a discussion and application of likelihood ratio functions to mineral potential mapping). Estimate $\lambda(x : x_1, \dots, x_m)$ at every pixel. According to the model, the pixel with the largest estimated value is considered to be the location

most likely affected by future landslides. Using the estimate value of the favorability function, a predicted hazard map showing the relative hazard level on a continuous scale is generated at every pixel in the map.

4.1. Simultaneous use of thematic and continuous data layers

The likelihood ratio function not only satisfies the first three conditions of the favorability function but also can satisfy the fourth condition of simultaneously using thematic and continuous data layers, if an additional assumption is made. First, at pixel x , simplify the notation by assuming that the first k layers correspond to thematic data layers and the remaining h layers represent continuous data layers. Then let $f\{x_1, \dots, x_k, y_1, \dots, y_h|M\}$ and $f\{x_1, \dots, x_k, y_1, \dots, y_h|\bar{M}\}$ be the multivariate frequency distribution functions from M , and $\bar{M}$, respectively, where the first k values, $x_1, \dots, x_k$ correspond to categorical data layers and subsequent h values, $y_1, \dots, y_h$ represent continuous data layers. Then the multivariate likelihood ratio function in Eq. (1) is denoted by:

$$\lambda(x : x_1, \dots, x_k, y_1, \dots, y_h) = \frac{f\{x_1, \dots, x_k, y_1, \dots, y_h|M\}}{f\{x_1, \dots, x_k, y_1, \dots, y_h|\bar{M}\}}. \quad (2)$$

To handle the multivariate frequency distribution functions of two different types of data layers, the following assumption of conditional independence is necessary:

$$f\{x_1, \dots, x_k, y_1, \dots, y_h|M\} = f\{x_1, \dots, x_k|M\} \times f\{y_1, \dots, y_h|M\},$$

and

$$f\{x_1, \dots, x_k, y_1, \dots, y_h|\bar{M}\} = f\{x_1, \dots, x_k|\bar{M}\} \times f\{y_1, \dots, y_h|\bar{M}\}. \quad (3)$$

Under assumption in Eq. (3), the $k + h (= m)$ dimensional multivariate frequency distribution function, $f\{x_1, \dots, x_k, y_1, \dots, y_h|M\}$ is expressed as a multiple of the k -dimensional multivariate discrete distribution function for categorical data and the h -dimensional multivariate continuous distribution function for continuous data. The same argument is applied to $f\{x_1, \dots, x_k, y_1, \dots, y_h|\bar{M}\}$. Under Eq. (3), the likelihood ratio function is simply a multiple of

two likelihood ratio functions:

$$\lambda(x : x_1, \dots, x_k, y_1, \dots, y_h) = \lambda(x : x_1, \dots, x_k) \times \lambda(x : y_1, \dots, y_h), \quad (4)$$

Hence, the likelihood ratio function will be estimated as a multiple of two estimated likelihood ratio functions, one for categorical data layers and one for continuous data layers.

To illustrate the conditional independence of thematic and continuous data layers, consider a bedrock lithology layer and a slope angle layer. In most cases, it may be reasonable to assume that the slope angle in a pixel is not correlated with the particular lithologic unit in the pixel. If the two layers are assumed independent they are conditionally independent given either the area affected by landslides $\mathbf{M}$ or the remaining landslide-free area $\overline{\mathbf{M}}$. The relationship is “conditional” because consideration is restricted to only part of the study area, $\mathbf{M}$ or $\overline{\mathbf{M}}$, at a time.

In contrast, consider a “vegetation coverage” data layer and an elevation data layer for a large study area. In some situations, vegetation may systematically vary according to elevation. An assumption of conditional independence between the two layers is not reasonable. In the case study, we will see how prediction patterns are affected if the conditional independence assumption is violated.

4.2. Estimating the likelihood ratio function for thematic data layers

There are two procedures to estimate the likelihood ratio function for categorical data layers, $\lambda(x : x_1, \dots, x_k)$ for a pixel x with k pixel values $(x_1, \dots, x_k)$, each x_i representing a categorical class where the pixel x is located in the i th data layer. Each procedure requires a different assumption. The first procedure requires conditional independence as discussed in Eq. (3). Assume the k categorical layers are conditionally independent. Then,

$$\lambda(x : x_1, \dots, x_k) = \lambda(x : x_1) \dots \lambda(x : x_k). \quad (5)$$

Instead of considering all k layers at one time, the likelihood ratio function may be estimated as a multiple of k separate univariate likelihood ratio functions. Each univariate ratio function is estimated by an individual categorical data layer in conjunction with the distribution of

occurrences of landslides:

$$\check{\lambda}(x : x_i) = \frac{\text{\# of landslide pixels in } x_i \text{ category of the } i\text{th layer}}{\text{\# of non-landslide pixels in } x_i \text{ category of the } i\text{th layer}}. \quad (6)$$

Using Eq. (6) under the conditional independence assumption in Eq. (5) yields an estimate:

$$\check{\lambda}(x : x_1, \dots, x_k) = \check{\lambda}(x : x_1) \dots \check{\lambda}(x : x_k). \quad (7)$$

An alternative estimation procedure requires assuming that two distribution functions, $f\{x_1, \dots, x_k | \mathbf{M}\}$ and $f\{x_1, \dots, x_k | \overline{\mathbf{M}}\}$ for $\lambda(x : x_1, \dots, x_k)$ are the multivariate multinomial frequency distribution functions (Johnson and Kotz, 1969) from $\mathbf{M}$ and $\overline{\mathbf{M}}$, respectively. Then, the maximum likelihood estimates of these two multinomial distribution functions are the two corresponding $m_1 \times m_2 \times \dots \times m_k$ cross-classified contingency tables, one for $\mathbf{M}$ and the other for $\overline{\mathbf{M}}$. By taking the ratios of two corresponding cells in two contingency tables, we obtain the second estimate:

$$\lambda(x : x_1, \dots, x_k) = \frac{\text{\# of landslide pixels within } U_{x_1 \dots x_k}}{\text{\# of non-landslide pixels within } U_{x_1 \dots x_k}}, \quad (8)$$

where $U_{x_1 \dots x_k}$ is defined as the unique condition subarea (Chung et al., 1995; Clerici et al., 2002) with the values $x_1, \dots, x_k$, belonging to the x_1 categorical class in the 1st layer and the x_k class in the k th layer. The estimate in Eq. (8) is identical to estimators used in the “conditional analysis” by Clerici et al. (2002) when all layers were converted to categorical data.

Use of Eq. (8) is not recommended if the total number of unique condition subareas becomes large (e.g., the number of categorized data layers is more than three or/and the number of classes in each layer exceeds five). Whenever the number of unique condition subareas becomes large, the number of pixels in the unique condition subareas tends to become very small and a large number of the $\lambda(x : x_1, \dots, x_k)$ equal to zero. When zero is integrated with the likelihood ratio functions from the continuous data layers, pixels with $\lambda(x : x_1, \dots, x_k) = 0$ will become zero regardless the likelihood ratio values from the continuous data layers.

In Fig. 1(A), black and grey bars represent the empirical distribution functions of bedrock geology thematic units in landslide areas (687 pixels) and in

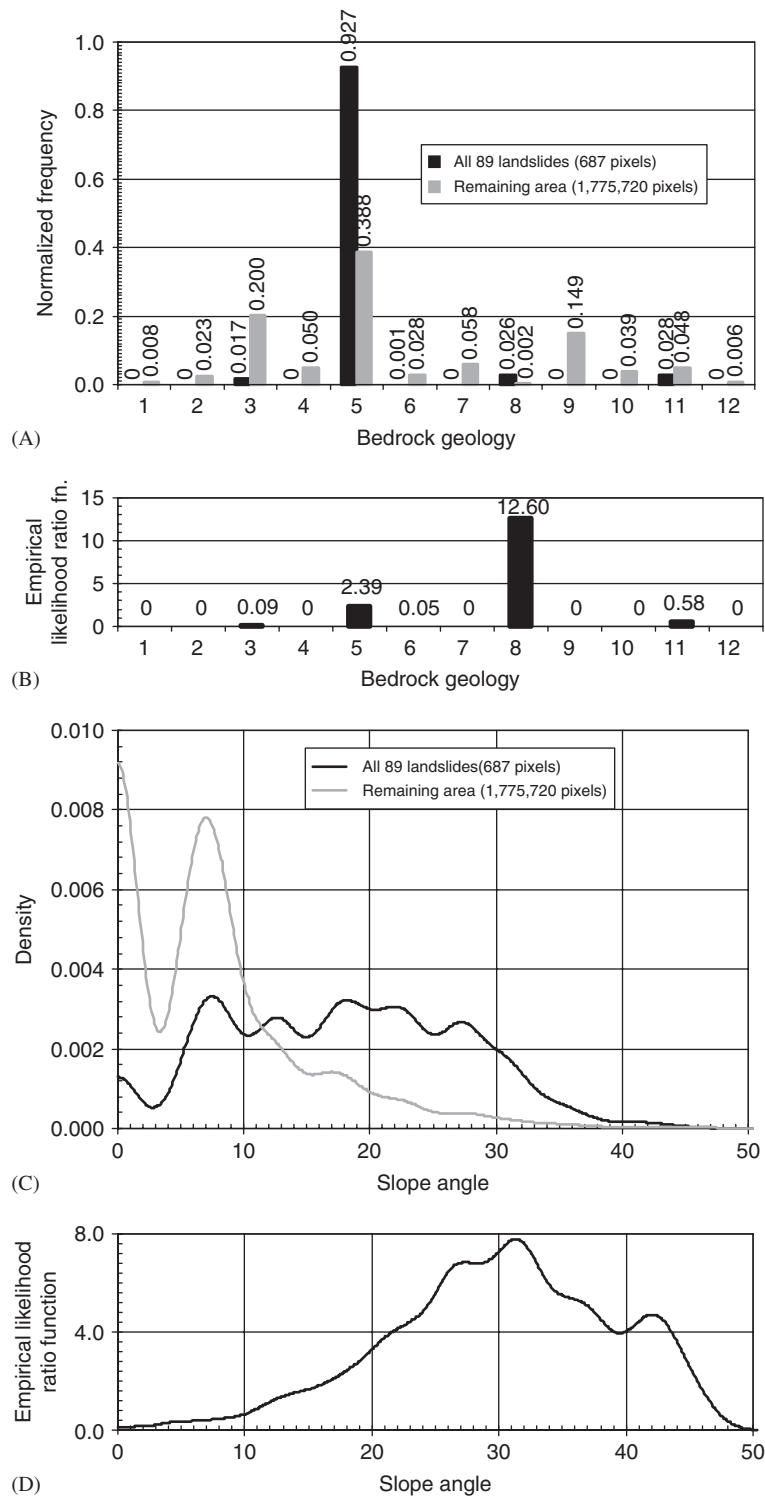


Fig. 1. Empirical distribution functions and likelihood ratios functions for La Baie study area, Canada. (A) Black and grey bars represent empirical distribution functions of Bedrock geology thematic units in landslide areas and in landslide-free areas. Labels on bars show frequencies of bedrock units. (B) Ratios of two empirical distribution functions in (A). (C) Black and grey bars represent empirical distribution functions of slope angles in landslide areas and in landslide-free areas. (D) Ratios of two empirical distribution functions in (C).

the remaining areas (1,775,720 pixels). Labels on the polygons show the percentage frequencies of bedrock units. Only five units (3, 5, 6, 8 and 11) host 1.7%, 92.7%, 0.1%, 2.6% and 2.8% of the landslides, respectively. Consequently, only five estimated ratios based on Eq. (6) are greater than 0. The remaining seven ratios are 0. The ratios of two empirical distribution functions illustrating the differences are shown in Fig. 1(B); labels give the ratios of the units. The ratios show that pixels in only two bedrock units, 5 (2.39 times) and 8 (12.6 times), are likely to be affected by landslides.

4.3. Resolution and scale problems with thematic data

When the original map from which a thematic data layer has been derived is based on observations that have relatively high resolution compared to pixel size, the likelihood ratio function in Eq. (7) provides reasonable estimates. In these conditions it is reasonable that hazard estimates may change abruptly across boundaries between two categorical classes. However, if the original map scale is relatively small compared to the pixel resolution, drastic changes at boundary lines are not appropriate. For example, suppose an original small-scale map, say 1:1,000,000, has been used to derive a data layer having pixels that represent 10×10 m areas. Then, a 1 mm thick boundary line on the original map represents a width of 1 km on the ground, or one hundred 10 m pixels. Drastic changes at such boundary lines are not appropriate. The scale of the map from which spatial information is obtained and the pixel resolution when the information is to be used must be considered in estimating the likelihood ratio function in Eq. (7). In particular, if the map scales of the original k thematic data layers are different, the estimation in Eq. (8) must explicitly address the different resolutions of the data.

Define β as the likely uncertainty of boundaries in terms of numbers of pixels. Obviously, β will be a function of the original map scale, pixel resolution and the mapmaker's subjective interpretation of the original data. The subjective component includes the numbers of factual observations needed to establish boundary lines (e.g., the number of outcrops necessary to construct a bedrock lithology map), and even the experience and skill of the field mapper. In the example of a 1:1,000,000 map and a derived data layer with a 10 m pixel resolution, β should be at least 50, implying that a line on a

1:1,000,000 map has a positional uncertainty of at least 1 mm. For any thematic classification map layer, β , termed *the radius of the likely uncertainty of boundaries in the layer*, plays a significant role when combining thematic data layers with different scales and resolutions.

If $\beta_i = 1$ for the i th thematic data layer, then $\tilde{\lambda}(x : x_i)$ will be used in Eq. (6) for the pixel x with the pixel value x_i . If $\beta_i > 1$, the estimator $\tilde{\lambda}(x : x_i)$ in Eq. (6) should be modified. Suppose that a circle with radius β_i drawn at the center of pixel x in the i th thematic data layer contains g pixel values, $x_{i1} \dots x_{ig}$. Of course, one of the $x_{i1} \dots x_{ig}$ must be x_i . Let $\rho_{i1} \dots \rho_{ig}$ be portions of the area occupied by the corresponding pixel values, $x_{i1} \dots x_{ig}$ within the circle. Then, instead of $\tilde{\lambda}(x : x_i)$ in Eq. (6) based on one pixel value, x_i at x , the likelihood ratio function for pixel x is given by:

$$\tilde{\lambda}(x : x_i) = \rho_{i1} \tilde{\lambda}(x : x_{i1}) + \dots + \rho_{ig} \tilde{\lambda}(x : x_{ig}), \quad (9)$$

where $\tilde{\lambda}(x : x_{i1}), \dots, \tilde{\lambda}(x : x_{ig})$ are obtained from Eq. (6).

The procedure is illustrated in Fig. 2, for pixel values x_{ip} , x_{iq} , x_{ir} , and x_{is} which occupy proportions

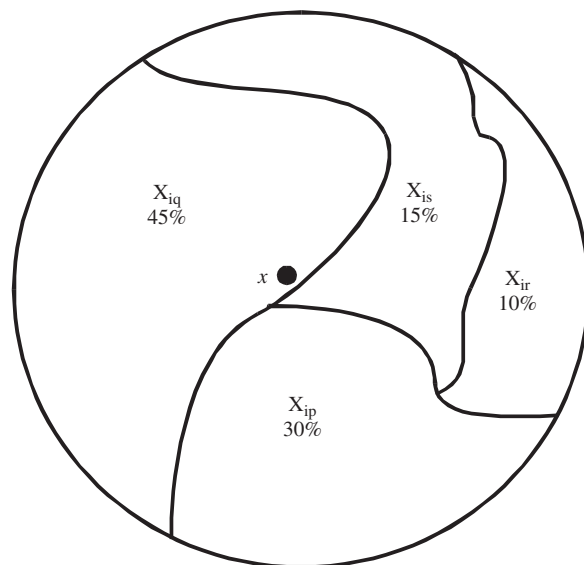


Fig. 2. Schematic expression of likely uncertainty associated with thematic unit boundaries due to map scale and pixel resolution relationships, when radius of likely uncertainty of boundaries in layer is larger than 1. Pixel values X_{ip} , X_{iq} , X_{ir} , and X_{is} are for pixels within circle surrounding pixel x with radius. Relative % refers to portion of areas occupied by corresponding pixel values, X_{ip} , X_{iq} , X_{ir} , and X_{is} within circle. Instead of $\tilde{\lambda}(x : x_i)$ in (6) based on one pixel value, X_{iq} at x , likelihood ratio function for pixel x is given by $\tilde{\lambda}(x : x_i) = \rho_{i1} \tilde{\lambda}(x : x_{i1}) + \dots + \rho_{ig} \tilde{\lambda}(x : x_{ig})$ in (9). See text for explanation.

of 0.30, 0.45, 0.10 and 0.15 of the circle around pixel x . At pixel x itself, the value is x_{iq} . If $\beta_i = 1$, then $\tilde{\lambda}(x : x_i) = \tilde{\lambda}(x : x_{iq})$. However if $\beta_i > 1$, then it should be estimated by $\tilde{\lambda}(x : x_i) = 0.3\tilde{\lambda}(x : x_{ip}) + 0.45\tilde{\lambda}(x : x_{iq}) + 0.1\tilde{\lambda}(x : x_{ir}) + 0.15\tilde{\lambda}(x : x_{is})$ based on Eq. (9), instead of $\tilde{\lambda}(x : x_{iq})$. The prediction map shown in Fig. 5 for the La Baie study area was based on a bedrock geology data layer and a forest coverage data layer whose β 's were set to 10 because the original information came from 1:100,000 paper maps. On paper maps of that scale, 1 mm represents 100 m on the ground, equivalent to $20 \times 5 \times 5$ m pixels; consequently, the radius $\beta = 10$.

Using Eq. (9) repeatedly k times, one for each layer, we obtained $\tilde{\lambda}(x : x_1), \dots, \tilde{\lambda}(x : x_k)$. Under the assumption in Eq. (5), an estimate that depends on the map scales of the k thematic data layers and the pixel resolution is obtained by:

$$\tilde{\lambda}(x : x_1, \dots, x_k) = \tilde{\lambda}(x : x_1) \dots \tilde{\lambda}(x : x_k). \quad (10)$$

4.4. Estimating the likelihood ratio function for continuous data layers

Only three of many possible procedures are presented here for estimating the likelihood ratio function for continuous data layers, $\lambda(x : y_1, \dots, y_h)$ for a pixel x with h pixel values $(y_1, \dots, y_h)$, where each y_j represents a real value at pixel x in the j th continuous data layer; each of the three procedures require different assumptions. The first procedure requires conditional independence similar to Eq. (5). Assume that the h continuous layers are conditionally independent and hence:

$$\lambda(x : y_1, \dots, y_h) = \lambda(x : y_1) \dots \lambda(x : y_h). \quad (11)$$

Instead of considering h -dimensional multivariate distribution functions, the likelihood ratio function may be estimated as a multiple of h separate univariate likelihood ratio functions, where each univariate ratio function is estimated by:

$$\tilde{\lambda}(x : y_j) = \frac{\tilde{f}\{y_j|\mathbf{M}\}}{\tilde{f}\{y_j|\overline{\mathbf{M}}\}}, \quad (12)$$

where $\tilde{f}\{y_j|\mathbf{M}\}$ and $\tilde{f}\{y_j|\overline{\mathbf{M}}\}$ are two empirical distribution functions estimated from the j th continuous data layer in conjunction with the distribution of occurrences of landslides.

In Fig. 1(C), the black and grey curves represent empirical distribution functions of slope angles in

landslide areas (687 pixels) and remaining areas (1,775,720 pixels) for the La Baie case study. The empirical likelihood ratio function of slope angles is shown in Fig. 1(D). The two empirical distribution functions intersect at a slope angle of 12° . The empirical likelihood function in Fig. 1(D) suggests that pixels with slope angles between 12 and 45° are likely to be affected by landslides. The highest contrast (more than 4 times) between the two functions occurs in pixels with slope angles between 23 and 43° , implying that these pixels are most likely to be affected by landslides.

Using Eqs. (11) and (12) yields an estimate:

$$\tilde{\lambda}(x : y_1, \dots, y_h) = \tilde{\lambda}(x : y_1) \dots \tilde{\lambda}(x : y_h). \quad (13)$$

Another estimate of $\lambda(x : y_1, \dots, y_h)$ can be obtained by assuming the two corresponding distribution functions, $f\{y_1, \dots, y_h|\mathbf{M}\}$ and $f\{y_1, \dots, y_h|\overline{\mathbf{M}}\}$ are multivariate normal distribution functions, $N(\mu_{\mathbf{M}}, \Sigma_{\mathbf{M}})$ and $N(\mu_{\overline{\mathbf{M}}}, \Sigma_{\overline{\mathbf{M}}})$, where $\mu_{\mathbf{M}}$, $\Sigma_{\mathbf{M}}$, $\mu_{\overline{\mathbf{M}}}$ and $\Sigma_{\overline{\mathbf{M}}}$ are the h -dimensional mean vectors for $\mathbf{M}$, the $h \times h$ dimensional covariance matrix for $\mathbf{M}$, the h -dimensional mean vectors for $\overline{\mathbf{M}}$, and the $h \times h$ dimensional covariance matrix for $\overline{\mathbf{M}}$. Under an assumption of normality, a second estimate is obtained:

$$\lambda(x : y_1, \dots, y_h) = \frac{\sqrt{|\hat{\Sigma}_{\mathbf{M}}|} \exp\{-0.5(y - \hat{\mu}_{\mathbf{M}})' \hat{\Sigma}_{\mathbf{M}}^{-1} (y - \hat{\mu}_{\mathbf{M}})\}}{\sqrt{|\hat{\Sigma}_{\overline{\mathbf{M}}}|} \exp\{-0.5(y - \hat{\mu}_{\overline{\mathbf{M}}})' \hat{\Sigma}_{\overline{\mathbf{M}}}^{-1} (y - \hat{\mu}_{\overline{\mathbf{M}}})\}}, \quad (14)$$

where $y = (y_1, \dots, y_h)$ is an h -dimensional vector containing h observed values, one for each layer, for point x . $\hat{\mu}_{\mathbf{M}}$, $\hat{\Sigma}_{\mathbf{M}}$, $\hat{\mu}_{\overline{\mathbf{M}}}$ and $\hat{\Sigma}_{\overline{\mathbf{M}}}$ are sample mean vectors and sample covariance matrices for $\mu_{\mathbf{M}}$, $\Sigma_{\mathbf{M}}$, $\mu_{\overline{\mathbf{M}}}$ and $\Sigma_{\overline{\mathbf{M}}}$. Computational operations in Eq. (14) are vector and matrix calculations. The estimated likelihood ratio function is also referred to as linear discriminant function analysis with two different covariance functions (see Press, 1972). Computer programs to solve Eq. (14) are widely available (e.g., SPSS, 1994). If normality assumptions are reasonable, then the estimate in Eq. (14) is recommended. However, if the empirical distribution functions appear non-normal, then the following logistic estimator or the estimator shown in Eq. (13) under the assumption of conditional independence is recommended.

As discussed in Chung (1977, 1978), and Dai and Lee (2002), the logistic function can be used instead of Eq. (13). Assuming the probability that y is in $\mathbf{M}$

follows a logistic distribution function yields the equation:

$$\Pr\{\mathbf{M}|y_1, \dots, y_h\} = \frac{\exp(\alpha_0 + y_1\alpha_1 + \dots + y_h\alpha_h)}{1 + \exp(\alpha_0 + y_1\alpha_1 + \dots + y_h\alpha_h)}, \quad (15)$$

where $(\alpha_1, \dots, \alpha_h)$ are unknown parameters to be estimated from the data. Denote $\Pr\{\mathbf{M}\}$ as a prior probability that any pixel is in $\mathbf{M}$, the area of landslides. The likelihood ratio function is given by:

$$\begin{aligned} \lambda(x : y_1, \dots, y_h) &= \frac{\Pr\{\mathbf{M}|y_1, \dots, y_h\}}{1 - \Pr\{\mathbf{M}|y_1, \dots, y_h\}} \cdot \frac{1 - \Pr\{\mathbf{M}\}}{\Pr\{\mathbf{M}\}} \\ &= \frac{\Pr\{\mathbf{M}|y_1, \dots, y_h\}}{1 - \Pr\{\mathbf{M}|y_1, \dots, y_h\}} \cdot C \\ &= \exp\{\alpha_0 + y_1\alpha_1 + \dots + y_h\alpha_h\} \cdot C \end{aligned} \quad (16)$$

where $C = (1 - \Pr\{\mathbf{M}\})/\Pr\{\mathbf{M}\}$.

Using the maximum likelihood estimator $(\hat{\alpha}_0, \hat{\alpha}_1, \dots, \hat{\alpha}_h)$ for $(\alpha_0, \alpha_1, \dots, \alpha_h)$ in Eq. (16) (Cox, 1970) gives the estimate:

$$\hat{\lambda}(x : y_1, \dots, y_h) = \exp\{\hat{\alpha}_0 + y_1\hat{\alpha}_1 + \dots + y_h\hat{\alpha}_h\} \cdot \hat{C} \quad (17)$$

where $\hat{C}$ is an estimate of $C = (1 - \Pr\{\mathbf{M}\})/\Pr\{\mathbf{M}\}$.

For the La Baie study area, both the estimators in Eqs. (13) and (17) were obtained for three DEM continuous data layers, aspect angles, elevations and slope angles of the pixels. The estimators $(\hat{\alpha}_0, \hat{\alpha}_1, \dots, \hat{\alpha}_h)$ for constant term, aspect angle, elevation, and slope angle were computed as $(-7.76800, 0.00149, 0.00760, 0.04716)$. As expected, the largest contributing factor for occurrences of landslides is the slope angle in a pixel.

4.5. Map scale problems in continuous data layers

If the original maps or images from which the continuous data layers are derived have different scales and resolutions, these differences must be considered in the estimation procedures of Eqs. (13), (14) and (17). For example, if two original images have resolutions of 10 and 30 m, and the pixels used in analyses are less than 10×10 m, then the difference in resolution plays a significant role in estimating the likelihood ratio function. The difference can be incorporated in the procedures used to obtain the empirical distribution functions of Eqs. (12), (13).

The kernel method is often used to estimate the unknown frequency distribution function of an

observed variable. The range of the variable is not divided explicitly into equal intervals as in a histogram; instead, pixel values are ordered across their range. The ordered sequence is then convolved with a kernel or weighting function, which may be a rectangular, sinusoidal, or Gaussian (normal) distribution in shape. The result, when standardized so the total frequency is unity, is the empirical frequency distribution function of the data set. The shape of the distribution resulting from the kernel method depends on the type and width of the kernel. The wider the kernel, the smoother the final distribution and the wider the distribution will be. The width of the kernel is termed its “spread” parameter.

As an example, Fig. 1(C) shows the empirical distribution functions of slope angles from landslide areas and remaining areas using Gaussian kernels with a spread parameter of 2% of the range of the slope angles in the database, and the corresponding likelihood ratio functions for the La Baie case study. Three additional sets of empirical distributions and corresponding ratio functions were generated using Gaussian kernels with parameters of 1, 4, and 8, as shown in Fig. 3(A). As the parameters increase, the empirical distribution functions become smoother but in some situations the ratio functions are less distinguishable for the two distributions. In practice, if a large spread parameter is used in a data layer, the layer will not play a significant role in prediction.

Rather than estimating the likelihood ratio by Eq. (12), the following estimator can be used:

$$\check{\lambda}_{\gamma i}(x : y_i) = \frac{\hat{f}_{\gamma i}\{y_i|\mathbf{M}\}}{\hat{f}_{\gamma i}\{y_i|\bar{\mathbf{M}}\}}, \quad (18)$$

where $\hat{f}_{\gamma i}\{y_i|\mathbf{M}\}$ and $\hat{f}_{\gamma i}\{y_i|\bar{\mathbf{M}}\}$ are the empirical distribution functions for $\mathbf{M}$ and $\bar{\mathbf{M}}$ areas for the i th layer using the spread parameter, γi . Using Eq. (18) instead of Eq. (12) gives:

$$\check{\lambda}(x : y_1, \dots, y_h) = \check{\lambda}_{\gamma 1}(x : y_1) \dots \check{\lambda}_{\gamma h}(x : y_h). \quad (19)$$

Different scales and resolutions can be incorporated in Eq. (12) at the stage of estimating $\hat{\Sigma}_{\mathbf{M}}$ and $\hat{\Sigma}_{\bar{\mathbf{M}}}$. However, such incorporation is somewhat subjective. It is also difficult and subjective to incorporate the difference into the logistic model in Eq. (17). Therefore, in practice, estimations in Eqs. (14) or (17) are not modified because of differences in scales or resolutions, as is done in Eqs. (19) and (13).

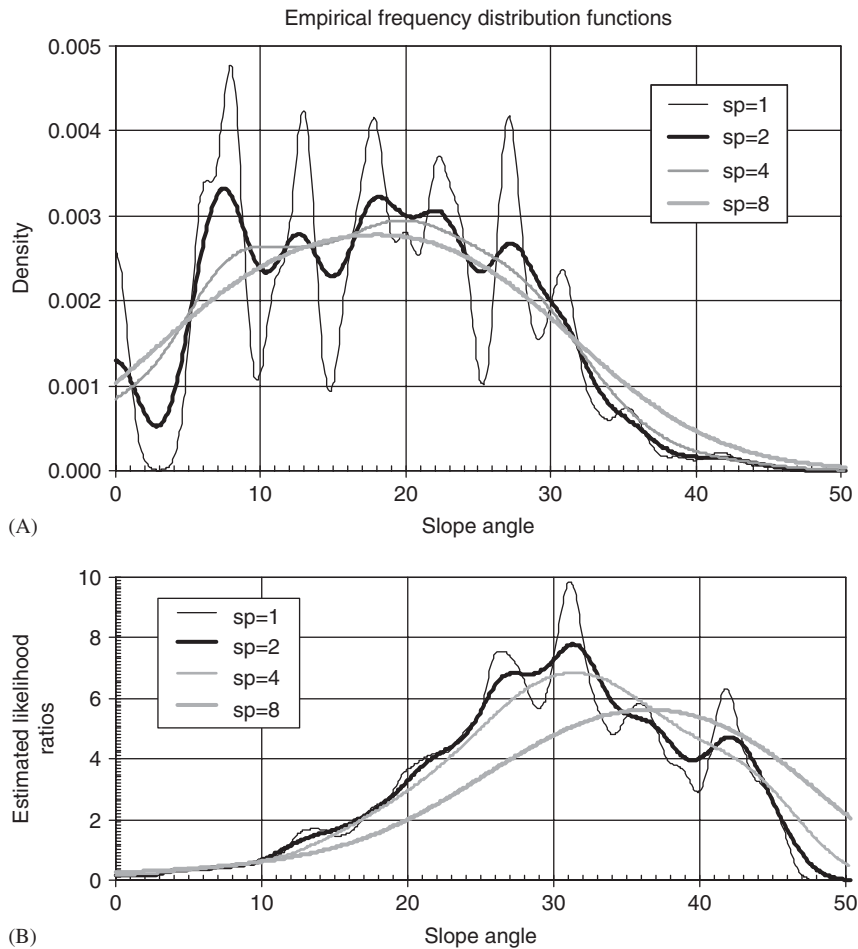


Fig. 3. (A) Empirical frequency distribution functions of slope angles in La Baie study area, Canada, using a Gaussian kernel with spread functions 1, 2, 4 and 8% of range. Normalized frequency values are on vertical axis. (B) Corresponding likelihood ratios functions estimated from empirical distribution functions in (A).

5. Combination procedure for generating relative hazard maps

Consider a pixel x with the first k pixel values, $x_1, \dots, x_k$, corresponding to categorical data layers and the following h pixel values, $y_1, \dots, y_h$, representing continuous data layers. Let $\beta_1, \dots, \beta_k$ be the corresponding radius of the likely uncertainty in boundaries of the k thematic layers and let $\gamma_1, \dots, \gamma_h$ be the corresponding likely uncertainty of pixel values in the h continuous layers.

The final estimate for $\lambda(x : x_1, \dots, x_k, y_1, \dots, y_h)$ is given by:

$$\hat{\lambda}(x : x_1, \dots, x_k, y_1, \dots, y_h) = \hat{\lambda}(x : x_1, \dots, x_k) \times \hat{\lambda}(x : y_1, \dots, y_h). \quad (20)$$

where $\hat{\lambda}(x : x_1, \dots, x_k)$ is obtained from Eq. (10) and $\hat{\lambda}(x : y_1, \dots, y_h)$ is obtained from either Eqs. (14), (17) or (19).

5.1. Generating hazard classes for a prediction map

By computing the estimated likelihood ratio function in Eq. (20) at every pixel in the study area, a hazard level is estimated at every pixel. These computed values may range from 0 to infinity. Because they express relative levels of hazard, they can be replaced by ranks (or orders) instead of the actual values. In the study area there are 1,776,407 pixels. Using Eq. (20) results in 1,776,407 estimated values, one at each pixel. These 1,776,407 values are sorted in decreasing order and replaced by their rank, ranging from 1 to 1,776,407. Dividing each rank by the number of pixels, 1,776,407, standar-

dizes the ranks. The resulting standardized ranks range from $1/1,776,407$ to 1; with the pixel having the most hazardous predicted level assigned the value 1, and the pixel having the smallest predicted hazard level assigned a value of 0.000000563 ($= 1/1,776,407$). To illustrate the relationship between computed ratio values and corresponding ranks, a scatter plot, as in Fig. 4 where x -axis is rank and the y -axis is computed value; was constructed in the La Baie study. Among 1,776,407 pixels, there are 574,846 pixels with estimated likelihood ratio values less than 0.0001, occupying 32.36% of the study area, and only 475,841 pixels, occupying 26.79% of the area, with a ratio value greater than 1. The largest ratio value is 1,163.14, with a corresponding pixel rank of 1.

By “slicing” (i.e., setting threshold values) either the estimated values or their ranks, a set of hazard classes is obtained—because they have a one-to-one relationship, either computed values or the equivalent ranks can be used to generate the hazard classes (Chung and Fabbri, 2003). In this study, the ranks are sliced. After specifying the desired number of classes, say n , the ranks are sliced into the n half-open intervals, $(0, 1/n]$, $(1/n, 2/n]$, ..., $((n-1)/n, 1]$. (The notation “ $(a, b]$ ” indicates that “ a ” is not

included in the interval but “ b ” is contained in the interval.) If the rank of a pixel falls in the interval, $((n-i-1)/n, (n-i)/n]$, the pixel will be assigned to the i th class. The set of hazard classes has the following characteristics: (1) All n classes include the same number of pixels and therefore each class represents the same sized area on the ground, equal to $1/n$ of the entire study area; (2) the last n th class is the most hazardous; and (3) it is relatively simple to interpret the hazard classes. A map of the set of hazard classes constitutes a prediction map.

The likelihood ratio function in Eq. (20) via Eqs. (10) and (19) has been computed at every pixel in the La Baie study area using five geomorphic and topographic map layers and 67 polygons enclosing the 89 landslides. Both β 's for bedrock geology data layer and a forest coverage data layer were set as “10” for Eq. (10) and all three parameters for the Gaussian kernels were set as “2” for Eq. (19). A prediction map has been generated with 200 hazard classes based on ranks and is shown in Fig. 5.

To examine the effect of violation of the assumption of conditional independence among the input data layers was examined using the likelihood ratio model in Eq. (20) via Eqs. (10) and (19), the slope angle data layer was entered

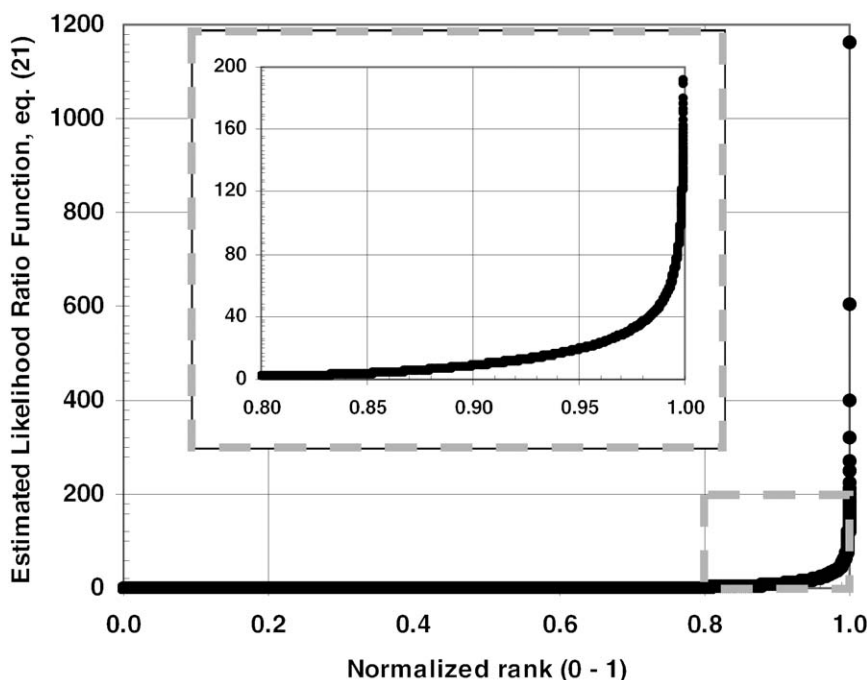


Fig. 4. Scatter plot of pair wise relationships in La Baie study area, Canada, between likelihood ratio values (vertical axis) and corresponding ranks (horizontal axis). Inset is enlargement of lower-right corner grey rectangle in figure.

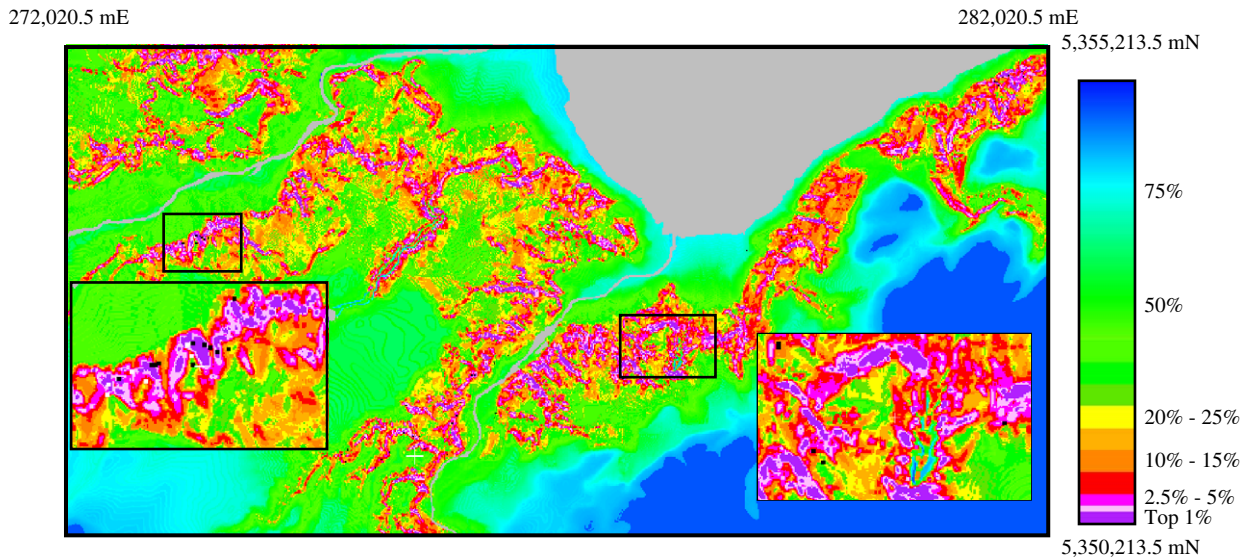


Fig. 5. Landslide hazard prediction map for La Baie study area, Canada. It was obtained by subdividing ranked values in 200 equally sized classes of 8882 pixels or 0.220 km^2 on ground (see Fig. 4). In two insets are two enlarged sub-areas, where locations of existing landslides are visible as black squares.

twice to simulate the use of two non-independent layers. Using six data layers instead of five data layers for Fig. 5, a prediction map of 200 hazard classes was generated. The new prediction map (six layers including slope angle layer twice) with Fig. 5 (five layers) show that over 94% of the most hazardous 10% areas were unchanged. The ranks of only 439 pixels (0.02% of the total area) were significantly altered. Similarly in a second experiment, the bedrock geology layer was entered twice and a prediction map with 200 hazard classes again generated. Compared with Fig. 5, over 98% of the most hazardous 10% areas were unchanged and the ranks of only 555 pixels differed by more than 10 ranks. Although the assumption of conditional independence was clearly violated in both experiments, prediction maps based on ranks were essentially unchanged, especially in the most hazardous classes. A reason for the robustness is that the likelihood ratios themselves are not interpreted; instead, the relative rankings of the predicted ratio values are used to predict the level of hazard. In practice, the degree of violation of conditional independence among input data layers would not be as severe as in these two experiments. Even though some degree of violation of the independence assumption is to be “expected” in real situations, these violations should not significantly alter rankings in the prediction maps.

6. Cross-validation procedure and estimation of conditional probability

A critical concern in landslide prediction modeling is interpreting the hazard classes on the prediction map. For this purpose, prediction-rate tables or graphical curves can be generated. In the procedure proposed here, occurrences of past landslides in the study area are divided into two mutually exclusive *Estimation* and *Validation* groups (Chung and Fabbri, 2003). Occurrences in the earlier time period of the estimation group are used to build a prediction model that generates a prediction map of hazard classes. A count of landslides in the later time period of the validation group that fall into hazard classes of the prediction map yields a prediction-rate table. In the La Baie case study, the estimation group consists of 25 landslides (222 pixels in 22 polygons) that occurred in 1964, the validation group consists of 64 post-1965 landslides (534 pixels in 55 polygons). As in Fig. 5, the likelihood ratio function in Eq. (20) via Eqs. (10), (19) has been computed at every pixel using five geomorphic and topographic map layers but only the 25 landslides in the estimation group. The resulting prediction map (not shown here) contains 200 hazard classes based on ranks. The 55 polygons representing the 64 landslides in the validation group were plotted on this prediction

map and the number of polygons that fell into each of the 200 hazard classes was counted. When a polygon is partially fell into a hazard class, at least 50% of the pixels in a polygon must be included in the class to be counted. The counts are weighted by the numbers of pixels in the polygons. Weighted counts form the “prediction-rate table” and the second column in Table 1 shows the counts for the 20 most hazardous classes. The same information is shown graphically in Fig. 6(A), with the horizontal axis representing the predicted hazard classes and the vertical axis showing the corresponding proportion of validation group landslides falling within the classes. Because they are based on ranks, the 200 classes are identical in size (8882 pixels covering 0.222 km² on the ground). Based on the prediction-

rate table, the conditional probabilities of future occurrences of landslides are estimated for each hazard class. In Table 1, class “200” is estimated as the most hazardous one. The next hazardous class is class “199”, and so on.

Ideally the weighted counts in the prediction-rate table (the 2nd column in Table 1, and in Fig. 6(A)) should be monotonic decreasing values. The larger index such as “200” shown in the first column in Table 1 is the hazardous class, the larger is the number of the landslides in the validation group that is expected to fall within the class. In the prediction-rate table, such a monotonic property is not maintained. For example, when class “191” (it contains no landslides in the validation group) is compared to the less hazardous class “183” (that contains 4.94% landslides of the validation group), the monotonic property is violated in the prediction-rate table. The violations are mainly due to having too many prediction classes. As shown in Fig. 6(B), by regrouping the hazard classes into a smaller number of classes, the monotonic property can be achieved in the case study (the most hazardous 50 classes were reduced into only five classes). The final number of classes is dependent on the effectiveness of the prediction model and/or the input data, and there is no unique way regrouping the hazard classes. Another possible reason for the violation of the monotonic property is that the prediction model and/or the input data are not sufficient to obtain reliable prediction results. In this case, the final number of classes is reduced to very few of them, usually two to four predicted hazard classes covering the entire study area. In the case study, having only five monotonic classes for the most hazardous 50 classes suggests that only a very limited success has been obtained by the prediction of future landslides.

If a time division is not possible because there is no information on time of occurrences of past landslides, the landslides can be randomly or spatially divided into two groups. In a random or spatial subdivision, the values computed are not natural extensions of the estimated probabilities of occurrence of future landslides. Extreme care must be given to the interpretation of prediction-rate tables generated by non-temporal partitions. The partitioning criteria for past landslides determine the interpretation of the prediction rate curve and the conditional probability. The prediction rate curves also can be used to measure the effectiveness of combinations of causal factors, the effectiveness

Table 1

Prediction-rate (column 2) and estimated probability (column 3) for 20 most hazardous classes of 200 classes shown in Fig. 5 for La Baie study area, Canada. Prediction rate was obtained by counting 55 polygons covering 64 post-1965 landslides from 200 hazard classes based on 25 landslides that occurred in 1964. Counting was weighted by number of pixels in polygons. Each hazard class consists of 8882 pixels covering 0.5% (0.222 km²) of study area. Estimated probability represents probability that a pixel in each class will be affected by a future landslide in next 40 years (2005–2044). It was obtained using values in column 2 and Eq. (21). Estimated probabilities were generated under assumption that 500 pixels will be damaged by future landslides in next 40 years

Hazard classes	Weighted count of 55 polygons for 64 post-1965 landslides	Estimated probability based on Eq. (21)
200	0.0952	0.00562
199	0.0550	0.00318
198	0.0330	0.00189
197	0.0164	0.00093
196	0.0165	0.00094
195	0.0385	0.00221
194	0.0165	0.00094
193	0.0888	0.00522
192	0.0559	0.00323
191	0.0000	0.00000
190	0.0339	0.00194
189	0.0338	0.00194
188	0.0083	0.00047
187	0.0083	0.00047
186	0.0055	0.00031
185	0.0055	0.00031
184	0.0055	0.00031
183	0.0494	0.00285
182	0.0165	0.00094
181	0.0165	0.00094

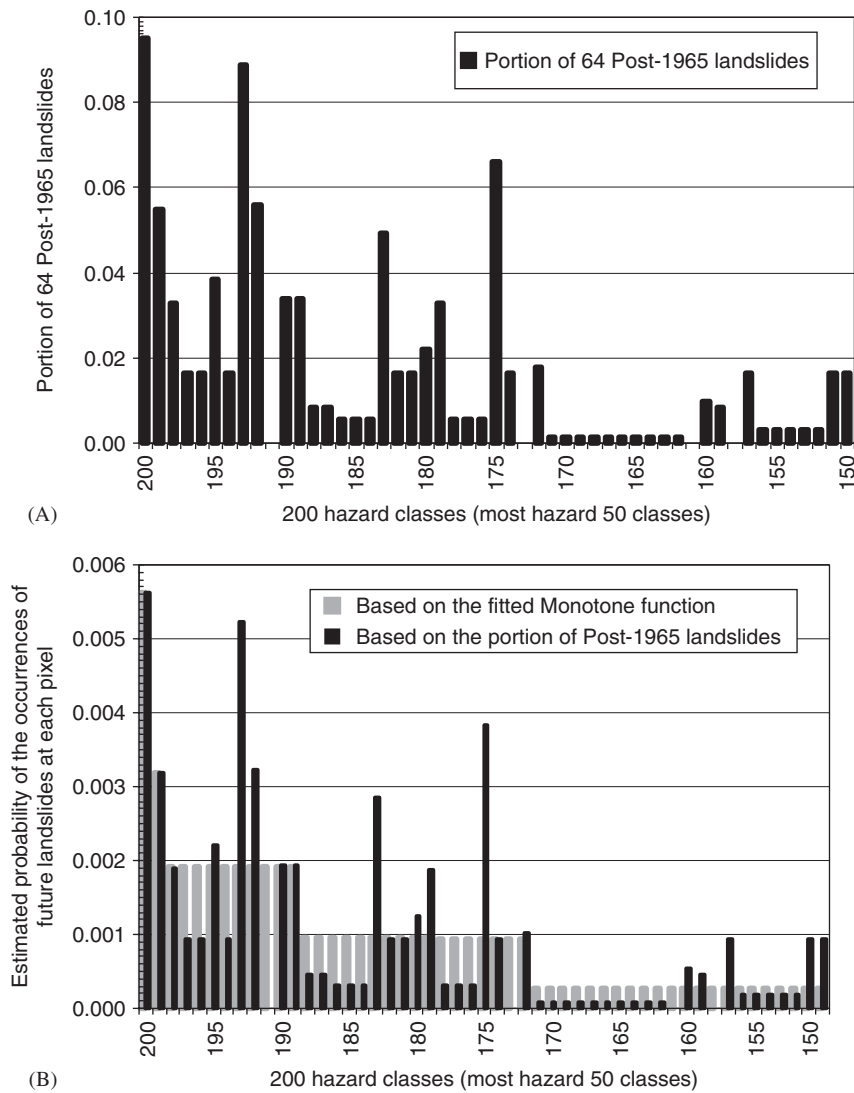


Fig. 6. (A) Graphical representation of prediction rate table shown in second column of Table 1. It is based on 200 hazard classes from likelihood ratio function in (20) via (10) and (19) computed at every pixel using two geomorphic (with both parameters, β are set to 10) and three topographic map layers (with all three parameters are set to 2) but only 25 landslides in estimation group. Number of 55 polygons representing 64 landslides in validation group that fell into each of 200 hazard classes was counted. Counts are weighted by numbers of pixels in polygons. Vertical axis is showing proportion of validation group landslides falling within each class. (B) Estimated empirical probability of occurrence of landslides in La Baie study area, Canada, using values in third column of Table 1.

of different favorability function models, and the temporal or spatial robustness of the predictions (Chung and Fabbri, 1999; Chung et al., 2002b).

Estimating the conditional probability of occurrence of future landslides in each pixel of a hazard class using the prediction rate table requires two additional parameters. First is the number of pixels in the class (or the area on the ground corresponding to the class) and the other one is the expected frequency and size of future landslides in the study area. In each hazard class, let n_b be the number of

pixels in the hazard class, let n_a be the number of pixels in the expected in future landslides, and let p_c be the portion of landslides in the validation group in the hazard class, obtained from the prediction-rate table. Then, the probability that a pixel, x , in the hazard class will be damaged by a future landslide is estimated by:

$$\hat{p}_x = 1 - [1 - p_c]^{n_a/n_b}. \quad (21)$$

Suppose that we expect that the landslides occur with the same rate in the next 40 years as that in the

past 40 years. Under the assumption, we can estimate the probability that a landslide will affect a pixel in the highest hazard class within the next 40 years using the prediction rates in the second column in Table 1. Each of the 200 hazard classes in the prediction map of Fig. 5 consists of 8882 pixels, covering 0.5% (0.222 km²) of the study area. Over the past 40 years, 64 landslides occupying 546 pixels have occurred. Assuming the rate remains constant, approximately 500 pixels will be affected by future landslides in the next 40 years (2005–2044) within the La Baie study area. Substituting $p_c = 0.0952$ in the second column Table 1, $n_b = 8882$ and $n_a = 500$ into Eq. (21) gives an estimated probability $\hat{p}_x = 0.0056$ that a pixel in the highest hazard class (0.222 km²) will be affected by a future landslide within the next 40 years. This procedure can be used to generate the third column of Table 1 and the corresponding graph (Fig. 6(B)).

Instead of estimating the probability that an individual pixel will be the site of a landslide, the probability that a building (or subarea occupying w adjacent pixels) will suffer a landslide can be estimated. The estimated probability, $\hat{p}_w$, is given by:

$$\hat{p}_w = 1 - [1 - \hat{p}_x]^w. \quad (22)$$

The conditional probabilities given in Eqs. (21), (22) can be used for a “cost-benefit analysis.” Similar ideas to those expressed by Eqs. (21) and (22) are discussed in terms of mineral potential maps in Chung et al. (2002a).

7. Concluding remarks and future work

A critical part of landslide risk analysis is not only to identify hazard classes in a study area but also to estimate the probability of occurrence of future landslides within the hazard classes. Validation allows estimation of an empirical measure so that the relationship between a prediction rate table and predicted hazard classes can be transformed into probabilities of occurrences.

In addition to the likelihood ratio function in Eq. (20) via Eqs. (10), (19), and Eq. (20) via Eqs. (10), (17): (the logistic model) has been estimated for every pixel using the same two geomorphic and three topographic map layers. Again, the data were divided into an estimation group containing only 25 landslides and a validation group containing 64 landslides. Landslides in the validation group were plotted on the prediction map and the number of landslide polygons in each hazard class counted and the prediction rate table, not shown here, was generated. Data from the two prediction rate tables, the second column in Table 1 using Eqs. (10), (19) and a new table (the numerical values are not shown here) using Eqs. (10), (17), are cumulatively plotted in Fig. 7. The horizontal axis of Fig. 7 indicates the portion of the study area classified as hazardous. The vertical axis is the proportion of validation group landslides within the specified hazard class. The black curve represents estimates based on the second column in Table 1 and is located above the grey curve that is based on the new rate table based on Eq. (10) and the logistic model Eq. (17). This

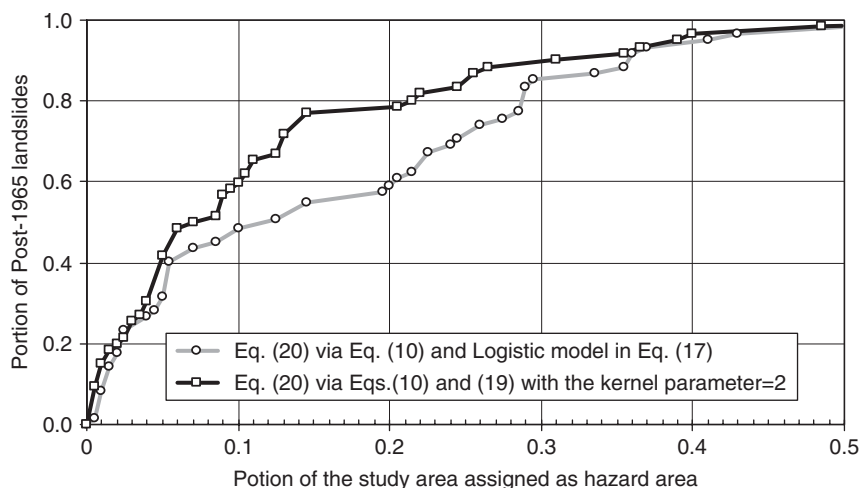


Fig. 7. Comparison of cumulative plots for Eq. (20) via Eqs. (10) and (19) and for Eq. (20) via Eq. (10) and logistic model in Eq. (17) used for landslide hazard prediction in La Baie study area, Canada.

means hazard classes from the Table 1; shown in Fig. 5, successfully predict more of the validation group landslides than do hazard classes created using the logistic model Eq. (17) in the La Baie study area.

Currently, research is underway to develop a methodology that can simultaneously incorporate several different resolutions or map scales into a single data layer. This problem arises whenever a study area consists of different subareas that were mapped by different geologists using different scales or resolutions.

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